What exactly is *clustering*?

Clustering is an **unsupervised** machine learning technique that groups similar data points together.

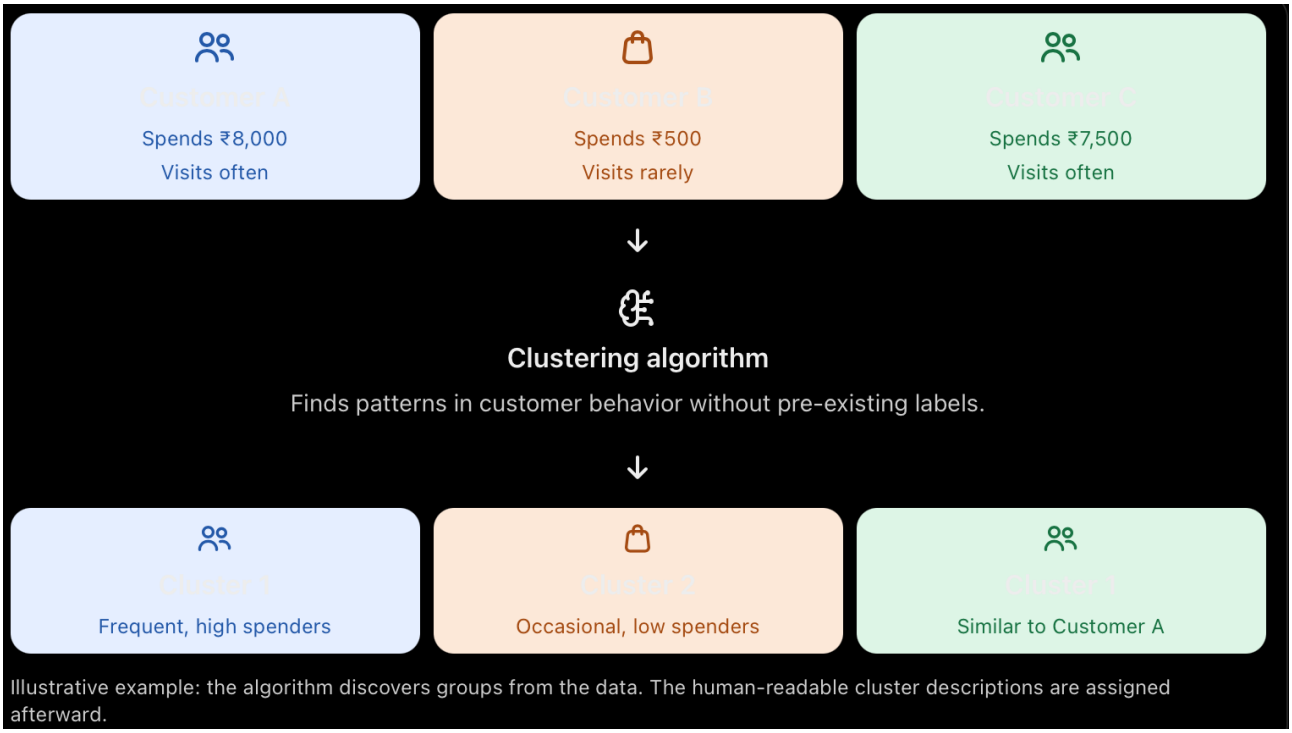


Imagine you own a mall. Thousands of customers visit every week. Some spend a lot of money, some visit frequently but buy inexpensive items, and some visit only occasionally.

You want to understand your customers so you can offer them better deals.

But there's a catch: you don't already know what type of customer each person is. You only have data about their behaviour.

That's where clustering comes in!



Imagine the mall records three things about each customer:

- Annual income: how much money they earn.
- Spending score: how much they tend to spend at the mall, rated from 1–100.
- Age: their age in years.

Here's a tiny sample dataset:

Customer	Annual income	Spending score
A	₹20,000	85
B	₹25,000	80
C	₹22,000	90
D	₹85,000	20
E	₹90,000	15
F	₹88,000	25

For simplicity, the income values are illustrative annual-income units; a real project must define its units clearly.

Notice how A, B, and C have similar spending scores and incomes, while D, E, and F have a different pattern. A clustering algorithm can group these customers based on their similarities.

The algorithm doesn't inherently know what a “budget-conscious shopper” or “premium customer” is. It finds the groups first; we interpret what they mean afterwards.

Why is clustering called *unsupervised* learning?

In supervised learning, you train using examples with known target labels. For instance, you might give a model customer data labelled “will purchase” or “will not purchase.”
In unsupervised learning, you give the algorithm data without predefined group labels and ask it to discover structure.

Supervised learning	Unsupervised learning
Learns from labeled examples	Learns from unlabeled data
Example: predict whether a customer will buy	Example: discover customer segments
Classification and regression	Clustering and dimensionality reduction

The four main reasons we need clustering

1. Customer segmentation

Group customers with similar behaviour so businesses can personalise offers, loyalty programmes, and marketing campaigns.

Mall example: Identify groups of frequent shoppers and occasional shoppers.

2. Discover hidden patterns

Find relationships or structures that aren't obvious from individual records.

Mall example: Discover a group of customers with high incomes but low spending scores.

3. Make better decisions

Use discovered groups to guide inventory planning, product recommendations, and business strategies.

Mall example: Understand which product categories different customer segments might prefer.

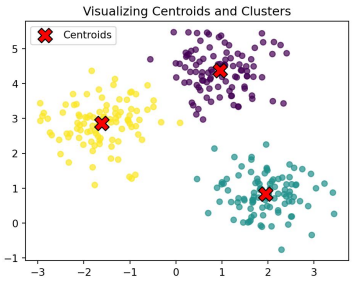
4. Organise complex data.

Clustering can also organise documents, group similar news articles, or help identify unusual patterns in network traffic.

Example beyond malls: Group similar documents without manually assigning a category to each one.

Clustering algorithms in Machine Learning

1. K-Means Clustering

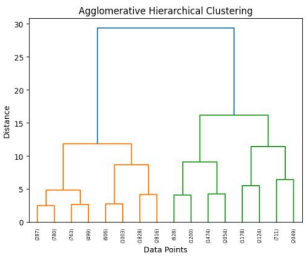


Groups data into K clusters by assigning points to their nearest centroid and repeatedly updating the centroids.

Best for: Well-separated, roughly round clusters.

Mall example: Segmenting customers based on annual income and spending score.

2. Hierarchical Clustering



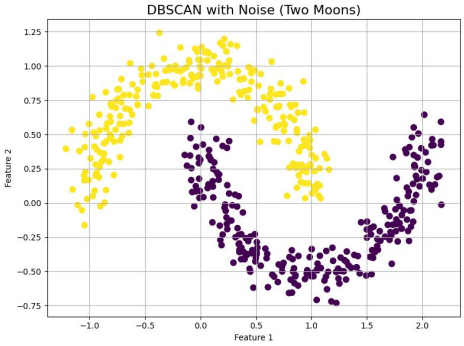
Builds a hierarchy of clusters, often by repeatedly merging smaller groups or splitting larger ones.

Best for: Understanding nested relationships between groups.

Mall example: First group customers broadly, then identify smaller customer segments.

Key concept: Dendrogram — a tree-like visualisation of cluster merges or splits.

3. DBSCAN



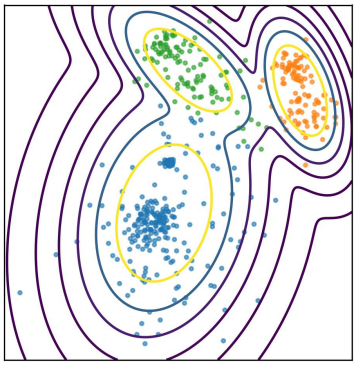
Groups points that form dense regions and marks isolated points as noise.

Best for: Irregularly shaped clusters and datasets with outliers.

Mall example: Discover dense groups of similar shopping behaviours without specifying the number of clusters beforehand.

Key concepts: `eps` and `min_samples`.

4. Gaussian Mixture Models (GMM)



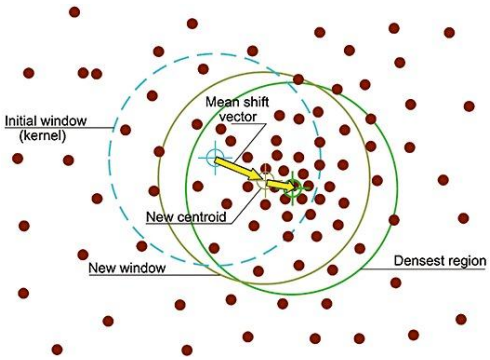
Models the data as a mixture of probability distributions, typically Gaussian distributions.

Best for: Overlapping or elliptical clusters.

Mall example: Estimate how likely a customer is to belong to each customer segment.

Key concept: Soft clustering — a customer can have probabilities of belonging to multiple clusters.

5. Mean Shift



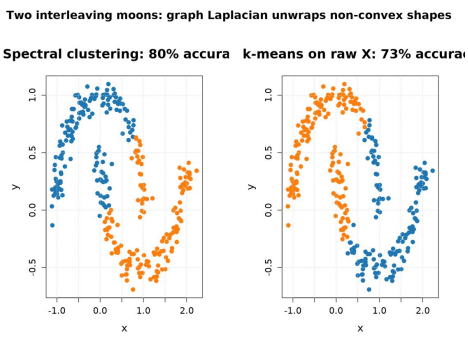
Finds dense regions by iteratively moving candidate centres towards areas with more data points.

Best for: Discovering clusters without specifying K in advance.

Mall example: Find naturally dense customer groups.

Limitation: Can be computationally expensive on large datasets.

6. Spectral Clustering



Uses similarities between data points and eigenvectors of a graph-based similarity matrix to identify groups.

Best for: Complex, non-convex cluster shapes.

Mall example: Separate customer groups that ordinary K-Means might struggle to distinguish.

Limitation: Can be expensive for large datasets.

The key takeaway

We need clustering because raw data tells us about individual records, while clustering can reveal the bigger patterns connecting them.